



A One-Stop Shop for Security Analysis and Learning

Computer security analysts have plenty of tools available nowadays to evaluate the security of corporate networks, servers and applications. These include Linux distributions specialising in security assessment, system recovery and digital forensic investigations. This article examines BackTrack, which stands out as the most comprehensive Linux distribution for security assessment of computer networks.

The security of computer networks and applications has gained tremendous importance over the last decade. Vulnerability assessment and penetration testing have now become routine tasks to evaluate the security of corporate networks, servers, Web applications, databases and other computer applications. Today, there are companies and professionals that specialise in the field of computer security analysis.

There are plenty of tools available to assist such professionals in various phases of security evaluation; most of these are Linux-based and are freely available. Such a wide variety of tools has led to the development of various security-oriented Linux distributions. These usually have a light-weight operating system based on the Linux kernel, a light-weight window manager for the GUI, and a collection of security tools of interest. They are mostly available in 'portable' form (as a live CD, DVD or USB), making it easy to use them without permanent installation on a computer. In addition to security assessment, many security-oriented Linux distributions also aid in system recovery and digital forensic investigations.

The most notable and comprehensive among these is BackTrack, which resulted from

the merger of two earlier Linux distributions, WHAX and Auditor Security Collection. Since its first public release (version 2 in late 2006), BackTrack has gained popularity rapidly, and attracted contributions from security professionals and open source developers. This makes it a premier and well-known tool-kit, with the richest collection of open source security tools in present times. The latest version, BackTrack 4, was released recently in January 2010.

BackTrack 4

In contrast to previous releases, which were based on Slackware, BackTrack 4 is based on Ubuntu (which, in turn, is based on Debian) and is a more complete, full-blown Linux distribution, instead of being just a live DVD. Powered by kernel version 2.6.30.9, it contains many bug fixes and improvements over previous releases. With BT4, standard applications and libraries are kept up to date by the Ubuntu repositories, while security tools are kept up to date by BackTrack repositories maintained by Offensive Security. After booting BT4, you can install new packages, upgrades or updates using the *apt-get* utility. This over-the-network-upgrade functionality was missing in previous BackTrack releases.